Totem pole PFC Converter-II

Introduction

In the realm of power electronics, efficient and reliable power conversion is paramount, especially in applications like electric vehicle (EV) charging systems. This lecture delves deeper into the Totem Pole Power Factor Corrector (PFC) converter, a sophisticated derivative of the bridgeless PFC topology. Unlike traditional PFC circuits that rely on a diode bridge rectifier at the input stage, the Totem Pole PFC eliminates this component, leading to enhanced efficiency and reduced losses. Understanding its operational nuances, control strategies, and inherent challenges is crucial for engineers designing high-performance power converters. This session builds upon previous discussions, focusing on the closed-loop control mechanisms and the critical issue of zero-crossing distortion, which significantly impacts the converter's performance.

Core Concepts

Term	Definition	Significance
`Totem Pole PFC`	A bridgeless PFC topology that eliminates the input diode bridge rectifier, using active switches for rectification.	Reduces conduction losses and improves efficiency compared to traditional PFCs.
`High Switching Frequency Leg`	Comprises switches S1 and S2, operating at a frequency (fsw) much higher than the line frequency (fs).	Responsible for the primary power factor correction and voltage regulation.
`Low Switching Frequency Leg`	Comprises switches S3 and S4, operating at the line frequency (fs), switching only once per half-cycle.	Manages the direction of current flow through the inductor based on the input voltage polarity.
`Unity Power Factor`	The condition where the input current is perfectly in phase with the input voltage, minimizing reactive power.	A primary control objective for PFC converters to ensure efficient power draw from the grid.

Term	Definition	Significance
`Closed-Loop Control`	A control system that uses feedback from the output (voltage, current) to adjust the input, maintaining desired performance.	Essential for regulating output voltage and achieving unity power factor in dynamic conditions.
`Modulator Block`	A circuit or algorithm that generates the Pulse Width Modulation (PWM) signals for the converter switches.	Translates the control signal (duty ratio) into switching commands for the power devices.
`Duty Ratio (d)`	The fraction of a switching period during which a switch is turned on.	Directly controls the average voltage across the inductor and thus the current and output voltage.
`Zero Crossing Distortion (ZCD)`	Undesirable spikes or irregularities in the input current waveform near the zero crossings of the input voltage.	A significant disadvantage of Totem Pole PFC, impacting Total Harmonic Distortion (THD) and power factor.
`Inductor Current Spike`	A sudden, sharp increase or decrease in the inductor current, often occurring during transitions.	Caused by delays in switch turn-on/off, leading to high voltage across the inductor and impacting current quality.
`RMS Current`	The root mean square value of the current flowing through a component.	Important for selecting appropriate power devices and assessing conduction losses.

Detailed Analysis

1. Totem Pole PFC Architecture and Basic Operation

The Totem Pole PFC converter is fundamentally a derivative of the bridgeless PFC, distinguished by the absence of a diode bridge rectifier at its input. Instead, it employs a combination of fast-switching and slow-switching active devices. The

high-frequency leg, typically consisting of switches S1 and S2 (often MOSFETs with body diodes), operates at a switching frequency (fsw) significantly higher than the line frequency (fs). These switches are responsible for the primary boost conversion and power factor correction. Conversely, the low-frequency leg, comprising switches S3 and S4, operates at the line frequency, switching only once per half-cycle to steer the current. This hybrid switching approach aims to reduce conduction losses by minimizing the number of semiconductor devices in the current path at any given time.

The design of the inductor (L) and capacitor (C) values in a Totem Pole PFC is analogous to that of a conventional boost PFC converter. The capacitor value is determined by the load power (P_load), line frequency (fs), output voltage (V_o), and the allowable output voltage ripple (ΔV_o). The inductor value is calculated based on the output voltage (V_o), maximum allowable current ripple (ΔiL,max), and the switching frequency (fsw). A key characteristic of single-phase PFC converters drawing unity power factor current is the presence of a second-line harmonic component on the output capacitor voltage, riding over the DC output voltage.

2. Closed-Loop Control Strategy

The primary control objectives for the Totem Pole PFC are twofold: regulating the output DC voltage to a desired reference value and ensuring that the input current drawn from the source maintains a unity power factor. This is achieved through a closed-loop control system that typically employs an average current mode control strategy.

The control loop begins by comparing the sensed output voltage (V_o) with a reference voltage (V_ref). The error signal is then fed into a voltage controller, which generates an output signal representing the magnitude of the desired input current. This output is clamped to a maximum allowable current value, determined by the converter's device ratings, to prevent overcurrent conditions.

This clamped magnitude is then sent to a multiplier block. In contrast to some boost PFC implementations that use the modulus of a unit sine wave, the Totem Pole PFC typically multiplies this magnitude with a unit sine wave (sin ωt) that is in phase with the input grid voltage. The accurate generation of this unit sine wave is critical, as grid voltages can have frequency variations and harmonics. While advanced Phase-Locked Loops (PLLs) and filters are often used for precise synchronization, a simplified approach involves sensing the grid voltage, dividing it by its peak value, and using this as the unit sine wave. This multiplication generates the reference input current (i_s,ref), which is ideally sinusoidal and in phase with the grid voltage.

The actual input current (i_s), which is the same as the inductor current (i_L) in this topology, is sensed and compared with i_s,ref. The error signal is then processed by

a current controller, whose output is fed into a gain block (ki) and subsequently to the modulator block. The modulator block's role is to calculate the appropriate duty ratio (d) for the high-frequency switches (S1, S2) and to determine the switching states for the low-frequency switches (S3, S4) based on the polarity of the input voltage (V_s). Both the output voltage and input current are continuously fed back into their respective control loops to maintain regulation and power factor correction. The overall control structure and transfer functions (Gi(s), Gv(s)) are largely similar to those of a boost PFC, allowing for the application of similar controller design principles.

3. Modulator Block Operation and PWM Generation

The modulator block is responsible for translating the calculated duty ratio `d(t)` into the actual PWM signals for the four switches (S1, S2, S3, S4). It also needs to sense the input voltage `vs(t)` to determine its polarity.

The duty ratios for the high-frequency switches S1 and S2 are determined as follows:
* When `vs > 0` (positive half-cycle):
* `dS1(t) = 1 - d(t)`
* `dS2(t) = d(t)`
* When `vs < 0` (negative half-cycle):
* `dS1(t) = d(t)`
* `dS2(t) = 1 - d(t)`

The duty ratios for the low-frequency switches S3 and S4 are determined by the input voltage polarity:
* When `vs > 0`:
* `dS3(t) = 0` (S3 is off)
* `dS4(t) = 1` (S4 is on)
* When `vs < 0`:
* `dS3(t) = 1` (S3 is on)
* `dS4(t) = 0` (S4 is off)

These duty ratios (`dS1(t)` , `dS2(t)` , `dS3(t)` , `dS4(t)`) are then compared with a high-frequency carrier waveform (at fsw) to generate the respective PWM signals (PWMS1, PWMS2, PWMS3, PWMS4). These PWM signals are then sent to the gate drivers of the switches.

4. Disadvantages: Zero Crossing Distortion

Despite its advantages, the Totem Pole PFC converter suffers from a significant drawback: zero-crossing distortion (ZCD) in the input current. This distortion occurs primarily during the transitions of the input voltage from positive to negative or

negative to positive. The root cause lies in the non-ideal behavior and synchronization challenges of the switches, particularly the low-frequency leg (S3, S4) and their interaction with the high-frequency leg (S1, S2).

Consider the scenario where the input voltage `vs` transitions from negative to positive (point A in the waveform). At this point, the low-frequency switch S3 should turn off, and S4 should turn on. Simultaneously, the duty ratios of S1 and S2 abruptly change (e.g., dS1 from 1 to 0, dS2 from 0 to 1).

Scenario 1: Delay in S4 Turn-on
If there is a delay in turning on S4 (due to dead time, device characteristics, or control synchronization issues), S3 might have already turned off, leaving S4 off as well for a brief period. During this dead time, the body diode of S3 might conduct due to the inductor current. However, if S4 is delayed, the output capacitance of S4 can charge up to the output voltage `V_o`. With `vs` near zero, and S2 potentially on, the voltage across the inductor `vL` can become approximately `V_o`. Since `V_o` is typically high (e.g., 400V), this large positive voltage across the inductor (`vL = L * di/dt`) causes a sudden, steep positive slope in the inductor current, leading to a positive current spike.

Scenario 2: S4 is On, S1 is On for Short Duration
Once S4 successfully turns on, the voltage across the low-frequency leg is very small. If S2 is on for a longer duration (as `vs` becomes positive), the inductor voltage `vL` is very small. However, in a switching cycle, S1 must also turn on for a short duration. When S1 is on, the inductor voltage `vL` becomes `vs - V_o`. Since `vs` is still very close to zero at the zero crossing, `vL` becomes a large negative value (e.g., `0 - 400V = -400V`). This large negative voltage across the inductor causes a sudden, steep negative slope in the inductor current, resulting in a negative current spike.

These positive and negative spikes manifest as zero-crossing distortion in the input current waveform. Instead of a smooth sinusoidal current, the current crosses zero multiple times or exhibits significant deviations. This distortion severely impacts the converter's performance, leading to a higher Total Harmonic Distortion (THD) and a reduced power factor (e.g., 0.97-0.99 instead of unity). The problem is exacerbated at light load conditions, where the distortion can be comparable to the peak grid current, further degrading THD and power factor. Addressing this requires advanced techniques like soft turn-on methods.

5. Advantages of Totem Pole PFC

Despite the zero-crossing distortion challenge, the Totem Pole PFC offers several compelling advantages:
* **Fewer Devices in Conduction:** By eliminating the input diode bridge, the

number of semiconductor devices through which the current flows at any given time is reduced, leading to lower conduction losses.
* **Reduced Switching Losses:** While S1 and S2 switch at high frequency, S3 and S4 switch only at line frequency, contributing to overall reduced switching losses compared to topologies where all devices switch at high frequency.
* **Reduced Component Count:** The absence of the diode bridge simplifies the circuit and reduces the overall component count, potentially leading to a more compact and cost-effective solution.
* **High Efficiency:** The combination of reduced conduction and switching losses generally results in higher overall efficiency, which is critical for applications like EV charging.

Key Takeaways

- [ ] Totem Pole PFC is a bridgeless topology, eliminating the input diode bridge for improved efficiency.
- [ ] It uses a high-frequency switching leg (S1, S2) for PFC and a low-frequency switching leg (S3, S4) for current direction.
- [ ] Control objectives are output voltage regulation and achieving unity power factor.
- [ ] The closed-loop control uses voltage and current feedback, a voltage controller, a multiplier with a unit sine wave, and a current controller.
- [ ] The modulator block generates PWM signals for all four switches based on duty ratio and input voltage polarity.
- [ ] A major disadvantage is Zero Crossing Distortion (ZCD) in the input current.
- [ ] ZCD is caused by delays in switch transitions (especially S4) and abrupt changes in duty ratios near zero crossings.
- [ ] ZCD leads to positive and negative inductor current spikes, increasing THD and reducing the power factor.
- [ ] ZCD is particularly problematic at light load conditions.
- [ ] Advantages include fewer conducting devices, reduced losses, and lower component count.

Conclusion

The Totem Pole PFC converter represents a significant advancement in power factor correction, offering inherent advantages in efficiency and component count due to its bridgeless architecture. Its ability to achieve high power factor and regulate output voltage through a well-defined closed-loop control strategy makes it suitable

for demanding applications. However, the challenge of zero-crossing distortion, stemming from the non-ideal synchronization of switches and their parasitic elements, remains a critical design consideration. This distortion can degrade the input current quality, impacting THD and power factor, especially under light load conditions. Designers must carefully balance the efficiency benefits with the need to mitigate these distortions, often employing advanced soft-switching or control techniques. As power demands continue to grow, particularly in sectors like EV charging, understanding these trade-offs and implementing robust solutions for Totem Pole PFC converters will be essential for developing the next generation of high-performance power electronics. The subsequent lectures will explore alternative operating modes, such as discontinuous conduction mode, to further optimize PFC converter designs.